INCH-POUND
MIL-DTL-24249B(SH)
25 January 2019
SUPERSEDING
MIL-P-24249A(SH)
20 July 1992

DETAIL SPECIFICATION

# SYNTHETIC RESIN MATERIAL, CELLULAR POLYURETHANE, RIGID, VOID-FILLING, FOAM-IN-PLACE, LARGE-SCALE INSTALLATION OF

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a large-scale installation system of a void-filling synthetic resin material that, after mixing and installation, will form a rigid, foam-in-place-type cellular polyurethane foam.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM C273/C273M | - | Standard Test Method for Shear Properties of Sandwich Core Materials |
| ASTM D1621 | - | Standard Test Method for Compressive Properties of Rigid Cellular Plastics |
| ASTM D1622/D1622M | - | Standard Test Method for Apparent Density of Rigid Cellular Plastics |
| ASTM D1623 | - | Standard Test Method for Tensile and Tensile Adhesion Properties of Rigid Cellular Plastics |
| ASTM D2126 | - | Standard Test Method for Response of Rigid Cellular Plastics to Thermal and Humid Aging |
| ASTM D2842 | - | Standard Test Method for Water Absorption of Rigid Cellular Plastics |
| ASTM D6226 | - | Standard Test Method for Open Cell Content of Rigid Cellular Plastics |
| ASTM C1304 | - | Standard Test Method for Assessing the Odor Emission of Thermal Insulation Materials |
| ASTM E84 | - | Standard Test Method for Surface Burning Characteristics of Building Materials |

(Copies of these documents are available online at www.astm.org.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Physical characteristics.

3.3.1 Ingredients. When mixed in accordance with the operating instructions of the automatic dispensing equipment specified (see 6.2), the ingredients shall form a rigid unicellular polyurethane foam for the intended application. The foam shall air-cure at ambient temperature, the minimum of which shall be as specified (see 6.2), and shall attain the physical properties specified in 6.5 (verified by the tests specified in 4.6.3 through 4.6.10) within 7 days after mixing.

3.3.2 Foamed material.

3.3.2.1 Uniformity per single pour. The cured material shall be essentially homogeneous (having uniform physical properties) throughout and firm in composition, with small, uniform cells. There shall be no large void openings (over ½ inch in any direction), large accumulations of unexpanded resin, or other occlusions (see 4.6.2). The automatic dispensing equipment used shall dispense foam at the rate specified (see 6.2).

3.3.2.2 Odor. The cured material shall be free from any objectionable odor, particularly residual isocyanate fumes, as specified in 4.9.

3.4 Physical property values. The foam material shall conform to the property values specified in 3.4.1 through 3.4.9.



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3.4.1 Density. The foamed block shall have a maximum density of 6.4 pounds per cubic foot (see 4.6.3).

3.4.2 Compressive strength. The compressive strength of the foamed block perpendicular to foam rise shall be greater than 60 but less than 130 pounds per square inch ($lb/in^2$), while the compressive strength parallel to foam rise shall be a minimum of 80 $lb/in^2$. Following both the humid aging test and the oil resistance test, compressive strength perpendicular to foam rise shall be greater than 60 but less than 130 $lb/in^2$ (see 4.6.4).

3.4.3 Tensile strength. The tensile strength of the foamed block perpendicular to foam rise shall be a minimum of 80 $lb/in^2$ (see 4.6.5).

3.4.4 Shear strength. The shear strength of the foamed block perpendicular to foam rise shall be a minimum of 50 $lb/in^2$ (see 4.6.6).

3.4.5 Unicellularity. The foam shall have a maximum open cell content of 15 percent (4.6.7).

3.4.6 Water absorption. The foam shall have a maximum water absorbance value of 0.10 pound per square foot of cut surface (see 4.6.8).

3.4.7 Oil resistance. The foam shall show no deterioration after being immersed in JP-5 fuel (see 4.6.9).

3.4.8 Humid aging. There shall be a maximum volume change of ±5 percent in the foam following humid aging (see 4.6.10).

3.4.9 Fire performance. When tested at a specimen thickness of 1 inch (nominal), the foam shall have a maximum flame spread index of 25 and a maximum smoke developed index of 250. There shall be no melting, dripping, or flaming droplets (see 4.6.11).

3.5 Toxicity and prohibited materials.

3.5.1 Toxicity. When evaluated in accordance with 4.7.1, the foam shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.7.1 and 6.4).

3.5.2 Prohibited materials. The foam shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the foam when a first article sample is required (see 3.1). This inspection shall include the tests of 4.6 through 4.8.

4.3 Conformance inspection. Conformance tests shall be performed by the installer at the site of installation. Tests for density (see 4.6.3) and compressive strength perpendicular to foam rise (see 4.6.4) shall be performed daily as required for large-scale installations (see 4.5.2). The tests shall be made on foam from each mixing head in use during the large-scale installation.

4.3.1 Equipment. Mixing equipment, metering pumps, hoses, dispensing equipment, and accessories shall duplicate expected shipboard operating conditions.

4.3.2 Sampling of mock-up pour. For purposes of first article mock-up pour demonstrations, a simulated compartment shall be constructed in accordance with 4.3.3.2.1. Foam rise shall be in the 14-foot direction (see 4.3.3.2.1). After a curing time of 24 hours, 20 samples shall be randomly cut out (an equal number from each layer) to provide 40 2¼- by 4- by 6-inch test specimens, each 2¼ by 4 by 6 inches; the dimension 2¼ inches shall be perpendicular to the direction of foam rise. Tests of density and compressive strength perpendicular to the direction to foam rise shall be made on each specimen (see 4.6.3 and 4.6.4).



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4.3.3 Sampling.

4.3.3.1 Sample preparation. Foaming and handling characteristics of the material shall be determined during the preparation of the test samples, as specified (see 6.2). Using automatic dispensing equipment, a quantity of foam ingredients for the sample molds of 4.5.3.1 and 4.5.3.2 shall be taken as applicable and then poured.

4.3.3.2 Sample molds.

4.3.3.2.1 Mock-up section for first article testing. Unless otherwise specified (see 6.2), the mock-up section shall be 4 by 8 by 14 feet. The mold requirements shall include the thickness of the material (steel), the size of the openings, the dimensions of the mold, and top or no-top configuration. The mold requirements shall also indicate when structural members, piping, or other simulated internal obstructions are required in the mock-up to obtain information about flow properties of the foam. The mock-up shall be able to be disassembled and an acceptable release agent (see 6.2) shall be used for ease of disassembly. Access to the mock-up section shall simulate the ship's construction.

4.3.3.2.2 Mock-up section for lot testing. A 1-$ft^3$ laboratory-prepared box pour shall be used for the lot test sample. The four walls and bottom shall be solid and shall be so constructed that the mold can be disassembled. A mold release agent shall be used to enable removal from the mold.

4.4 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall conform to the test conditions specified in 4.6.1.

4.5 Inspection lot. For sampling purposes, a lot shall consist of all components of a foam system, each manufactured as one batch but not necessarily delivered at one time.

4.5.1 Source inspection. From each lot (see 4.5), the installer shall take a quantity of foam resin material for the molds specified in 4.3.3.2.1. The mold samples shall be foamed, prepared, and tested as specified (see 4.6).

4.5.2 Conformance inspection preceding, during, and after installation in the large-scale voids. Prior to installation, concurrent with the pouring in the ship's compartments (approximately every 200 $ft^3$ of compartment space filled), and when pouring is concluded, the installer shall perform density and compressive strength perpendicular to the foam rise tests (see 4.6.3 and 4.6.4) using the mold specified (see 4.3.3.2.1). The tests shall be conducted at the installer's laboratory. Samples of the foam mix shall be poured into a 1-$ft^3$ mold. Each mold shall provide not less than two test specimens 2¼ by 4 by 6 inches (6 inches in direction of foam rise). Density and compressive strength perpendicular to the direction of foam rise shall be determined on each sample as rapidly as possible in order to provide information and guidance for any necessary adjustment of materials or mixing equipment. In the event that the laboratory results indicate that the foam does not meet the density and compressive strength requirements of this specification, all pouring shall immediately cease. The foam that has already been poured to fill the void shall be removed, the machinery shall be recalibrated, and the void shall be re-poured.

4.5.3 Preparation of test specimens.

4.5.3.1 Mock-up mold (see 4.3.3.2.1). Samples shall be selected in accordance with 4.3.2. The tests specified in 4.6 shall be conducted on each of these samples. The test specimens from these samples shall be prepared as specified (see 4.5.3.2 and 4.5.3.3).

4.5.3.2 Lot testing mold (see 4.3.3.2.2). After determining the density (see 4.6.3), one half of the sample block shall be cut into 1- or 2¼-inch-thick layers. The thickness of these layers shall be perpendicular to the direction of foam rise except for one 2¼-inch-thick layer, which shall be parallel. The other half shall be used to prepare tensile specimens. For the tensile specimens, the thickness shall be parallel to the direction of foam rise. Specimens shall be free of skin and shall be prepared in such a manner that the possibility of heating is minimized. Test specimens shall not be heated, immersed in water, or subjected to any mechanical or chemical treatment prior to testing, except as described herein.



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4.5.3.3 Size of specimens. The number and size of the specimens for each test as specified in 4.6 shall be prepared from the 1- or 2¼-inch-thick layers specified in 4.5.3.2. Unless otherwise specified herein, test specimens shall be selected at random from the prepared 1- and 2¼-inch-thick layers, as required. Prior to testing, each 1- by 4- by 4-inch specimen and 2¼- by 4- by 6-inch specimen shall be weighed to the nearest 0.1 gram, and each 1- by approximately 1.129-inch diameter specimen to the nearest 0.001 gram. The dimensions shall be measured to the nearest 0.01 inch and 0.001 gram and density calculated. Any specimen whose density is not within 5 percent of that of the sample block shall be rejected and replaced with a new specimen. Specimens for compressive strength and change in compressive strength after humid aging and oil immersion shall be cut from the same 2¼-inch-thick layers of foam.

4.6 Test procedures.

4.6.1 Test conditions. Unless otherwise specified (see 6.2), the atmospheric conditions surrounding the specimen prior to and during the test shall be 23±2 °C (73.4±3.6 °F), and 50±5 percent relative humidity. The temperature, the relative humidity, and the period of time for conditioning shall be recorded for each test. Results shall be recorded as the average value for the number of specimens tested, except for compressive strength, for which all individual values shall be recorded.

4.6.2 Uniformity in single pours. The test block molded in accordance with 4.3.3.2.2 shall be visually examined on all surfaces for any void openings, large accumulations of unexpanded resin, or any other occlusions. The test block shall be cut horizontally into three equal sections and the inside surfaces examined in a similar manner. Any void openings or large accumulations of unexpanded resin shall be cause for rejection (see 3.3.2.1).

4.6.3 Density. Irregular surfaces of the foam samples (such as the domed surface of the sample) shall be squared off to permit accurate measurement of dimensions. The density of the foam shall be determined in accordance with ASTM D1622/D1622M, except that one sample block shall be tested (see 3.4.1).

4.6.4 Compressive strength. Compressive strength shall be determined in accordance with procedure A of ASTM D1621, except that the record of average values does not apply and there shall be two 2¼- by 4- by 6-inch test specimens. Each individual value shall be recorded (see 6.5.2). Compressive strength perpendicular to foam rise shall be tested again following both the humid aging test and the oil resistance test and shall meet the requirements of 6.5.2.

4.6.5 Tensile strength. Tensile strength shall be determined in accordance with ASTM D1623 using a type A specimen, except that three specimens shall be tested (see 6.5.3).

4.6.6 Shear strength. Shear strength shall be determined in accordance with ASTM C273/C273M, except that three ½- by 2- by 6-inch specimens shall be tested (see 3.4.4).

4.6.7 Unicellularity. The percent open cell shall be determined in accordance with ASTM D6226, except that three approximately 1- by 1.129-inch diameter specimens shall be tested (see 3.4.5).

4.6.8 Water absorption. Water absorption shall be determined in accordance with ASTM D2842, except that the cell chord length determination shall be omitted and five 1- by 4- by 4-inch specimens shall be tested (see 3.4.6).

4.6.9 Oil resistance. The specimen shall be immersed in JP-5 turbine fuel conforming to MIL-DTL-5624. After 70 hours, the specimen shall be removed, lightly blotted with paper toweling, and compared to an unaged specimen of similar size for evidence of softening or dissolving (see 3.4.7). Three specimens approximately 1 by 1.129 inches in diameter shall be tested.

4.6.10 Humid aging. Humid aging shall be determined in accordance with ASTM D2126, except that five 2¼- by 4- by 6-inch specimens shall be tested (see 3.4.8).

4.6.11 Fire performance. The fire performance of the foam (see 3.4.9) shall be determined in accordance with ASTM E84.



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4.7 Toxicity and prohibited materials.

4.7.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.5.1 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the foam using data provided by the manufacturer/distributor to the NMCPHC (see 3.5.1 and 6.4).

4.7.2 Prohibited materials. Prohibited materials shall be verified as specified (see 3.5.2 and 6.2).

4.8 Lot acceptance/rejection criteria. If the new sample fails any of the tests specified in 4.6 during source inspection retest, the lot represented by the samples (original and new) shall be rejected (see 4.5).

4.9 Nose test for odor. The material shall produce no offensive odor when tested in accordance with ASTM C1304.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material is intended for shock absorption in large void-filling applications in Navy ships. Current stock of foam-in-place void-filling synthetic resin material may be used until depleted; future requisitions for foam-in-place void-filling synthetic resin material should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Total quantity of foam required.

c. When first article inspection is required (see 3.1).

d. The automatic dispensing equipment specified (see 3.3.1).

e. Minimum ambient temperature for air-cure, if applicable (see 3.3.1).

f. Requirements for an HHA (see 3.5.1 and 4.7.1).

g. Requirement for prohibited materials verification (see 3.5.2 and 4.7.2).

h. Where test pours are made, foam rate, depth per pour after rise, minimum time required between vertical successive pours, minimum time for test specimens to reach maximum physical properties, and quantity of foam in any single pour (see 3.3.2.1 and 4.3.3.1).

i. Foaming and handling characteristics (see 4.3.3.1).

j. Area of mock-up section from which samples are to be cut if other than specified (see 4.3.3.2.1).

k. Release agent to be used (see 4.3.3.2.1).

l. Inspection conditions, when other than specified (see 4.4).

m. Atmospheric conditions prior to and during the test, if other than specified (see 4.6.1).

n. Packaging requirements (see 5.1).

o. Requirements for the SDS and FAR clause 52.223-3 (see 6.3).

p. Ratios and tolerances for foam mixing and installation machinery (see 6.5.3).



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6.3 Safety data sheet (SDS). The contracting activity will be provided a Safety Data Sheet (SDS) at the time of contract award. The SDS will be provided in accordance with the requirements of FED-STD-313. The SDS will be included with each shipment of the material covered by this specification. Contracting officers must identify those activities requiring copies of SDSs. Additional required Government information is contained in FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract (see 6.2).

6.4 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.5 Precautions.

6.5.1 Ventilation. Isocyanate fumes may be present during foam application. As prescribed on the SDS (see 6.3), ventilation should keep vapor concentrations below acceptable limits. If ventilation as prescribed on the SDS cannot be provided, air masks should be worn.

6.5.2 Personal protective equipment. Anyone involved in foam application who may come in contact with isocyanates should wear chemical safety glasses or face shields, rubber gloves, and coveralls to protect against splattering.

6.5.3 Fire safety. During the mixing and installation of the foam, machinery should be calibrated frequently to ensure that system ingredients are being delivered according to the ratios and tolerances established (see 6.2). Hot work, including welding, burning, and use of any open flames, should be completed before any foam installation commences. In addition, smoking, electric heaters, or things producing a similar effect are prohibited from the site of installation and all surrounding areas.

6.6 Subject term (key word) listing.

Cellular plastic

Expanded plastic foam

Filler

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

## CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CQ

Preparing activity:
Navy – SH
(Project 9330-2018-009)

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